

Wireless Communication Exam Notes

EX 715 (BEI) / EX 751 (BEX) · IV-I · TU, IOE · tiers over 22 papers, 2070–2082 BS

2. Cellular Mobile Communication Concept

4 hrs · asked in **20 of 22 papers** · 5 syllabus sub-topics · heavy numerical load → see the Ch 2 numerical companion.

Read this first:

- Two questions carry the chapter. **Handoff** **TOP 8/22** and the **proof** $Q = \sqrt{3N}$ **TOP 6/22**. Bank both and most papers are already covered.
- Everything in this chapter hangs off **one trade-off**: small $N \rightarrow$ more capacity but worse SIR. Large $N \rightarrow$ better SIR but less capacity. Say that sentence in any answer here and it will fit.
- The examiner reliably pairs a theory part with a numerical part (**prove** $Q = \sqrt{3N}$, **then find optimal N for a given SIR**). Do not learn the proof without the numerical.

2.1 The cellular concept

What is the cellular concept? What is a cell footprint? Explain interference tier **1+4 5**

80 Ch · background for every other question in the chapter

The problem it solved

- Early mobile radio = **one high-power transmitter** on a tall tower.
 - Large coverage, but the same frequencies could **never** be reused inside it.
 - Bell's New York system supported **12 simultaneous calls over 1000 square miles**.
- Spectrum is fixed by the regulator → capacity could not grow.

Advantages of going cellular

- Low transmit power → smaller batteries, cheaper hardware, less interference.
- Frequency reuse → capacity grows w/o new spectrum.
- Capacity can be grown later by splitting cells, w/o redesigning the system.

The cellular idea (3 system-level ideas)

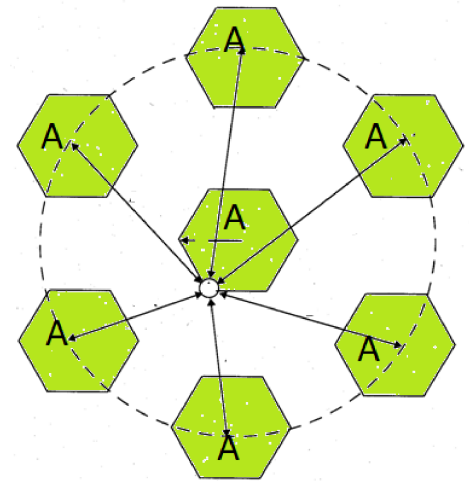
- Replace one high-power transmitter with **many low-power transmitters**, each covering a small area called a **cell**.
- Give **neighbouring cells different channel groups** → interference stays low.
- Reuse the same channel set** at cells far enough apart.
- Result: high capacity in a fixed spectrum, **with no new technology**.

Why a hexagon +Fig

- The ideal contour is a **circle**, but circles either **overlap** (double coverage) or leave **gaps** (no coverage).
- Only 3 regular shapes tile a plane w/o gaps or overlap: **triangle, square, hexagon**.
- For the same distance from centre to the farthest edge, the **hexagon covers the largest area** → **fewest cells, fewest base stations, lowest cost**.
- Hexagon is also the closest of the three to a circle → best match to a real omni-directional antenna pattern.
- It has exactly **6 equidistant neighbours**, and centre-to-centre lines are separated by multiples of 60° → makes the reuse geometry solvable.

Terms

- **Cell** = area covered by one cellular transmitter. The transmitter site itself = **cell site**.
- **Cell footprint** = the **actual** radio coverage of a cell.
 - Real footprints are irregular (terrain, buildings).
 - Irregular footprints are tolerable at first, but as cells are added they **break the reuse plan** and cause co-channel interference.
 - So a **regular shape is assumed** for planning.
- **Cluster** = the N cells that together use the complete channel set.
- **Co-channel cells** = cells using the same channel set as the target cell.
- **Interference tier** = a set of co-channel cells all at the **same distance** from the reference cell.
 - The nearest such set = **first tier**.
 - **In hexagonal geometry the first tier always has exactly 6 cells.**
 - Second and higher tiers are much further away, so their interference is normally neglected.



First tier: 6 co-channel cells, all at distance D from the reference cell.

2.2 Frequency reuse and cluster size

Explain the frequency reuse concept and the co-channel reuse ratio **6 4 3**

HOT 3/22 72 Ma, 74 Bh, 76 Bh · 74 Bh words it as “minimizing reuse distance maximizes spectral efficiency”

Frequency reuse / frequency planning

- Each BS gets a group of channels, used only inside its own cell.
- Neighbouring cells get **different** groups.
- Because coverage is confined to the cell, the **same group can be reused** at a cell far enough away.
- Design task = choose the reuse pattern so interference stays tolerable.

Capacity arithmetic (learn this, it is the base of every numerical)

- S = total duplex channels in the system.
- k = channels per cell.
- N = cells per cluster (cluster size).
- M = number of times the cluster is replicated.
- So $S = kN$, and system capacity

$$C = MkN = MS$$

- **Capacity is directly proportional to M** , the number of replications.
- **Frequency reuse factor** = $1/N$: each cell gets $1/N$ of the total channels.
- Typical $N = 4, 7, 12$.
- **Small $N \rightarrow$ cluster repeats more often $\rightarrow$ higher capacity**, but co-channel cells sit closer $\rightarrow$ worse interference.

Which N are allowed

- Hexagons only tile for

$$N = i^2 + ij + j^2 \quad i, j = 0, 1, 2, \dots$$

- Allowed N : 1, 3, 4, 7, 9, 12, 13, 16, 19, 21, ...
- Eg $i = 1, j = 1 \rightarrow N = 3$. $i = 2, j = 1 \rightarrow N = 7$.
 $i = 3, j = 2 \rightarrow N = 19$.

Channel assignment strategies syllabus, no PYQ yet

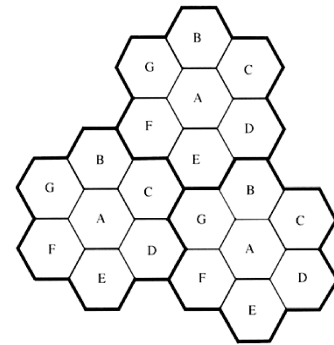


Figure 3.1 Illustration of the cellular frequency reuse concept. Cells with the same letter use the same set of frequencies.

Cells with the same letter use the same channel set.

Locating the co-channel cells + Fig

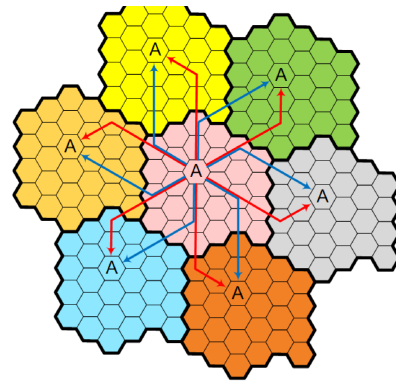
- From the centre of any cell:
 - move i cells perpendicular to one of the 6 faces,
 - turn 60° , move j cells.
- That lands on a co-channel cell. Repeating for all 6 faces gives the whole first tier.
- Clockwise and counter-clockwise turns give the two mirror routes.

Fixed channel assignment (FCA)

- Each cell gets a **predetermined** set of voice channels.
- A new call can only use an unused channel **of that cell**.
- If all of the cell's channels are busy, the call is **blocked**, even when a neighbour is idle.
- Simple, no signalling overhead, predictable.
- **Borrowing** variation: a busy cell may borrow a channel from a neighbour, supervised by the MSC, provided it does not break the reuse plan.

Dynamic channel assignment (DCA)

- Channels are **not** owned by any cell. They sit in a central pool at the MSC.
- On each request the MSC allocates a channel after checking
 - the candidate channel is unused in that cell,
 - and no co-channel cell within the reuse distance is using it.
- **Lowers blocking probability, raises capacity** because it follows the traffic.
- Cost: the MSC must collect occupancy and signal-strength data continuously → heavy storage and computation load.



$i = 4, j = 2$: the 6 nearest co-channel A cells.

2.3 Handoff

What is handoff? Explain the handover process and the types of handover with application 6 4+4 4

2+4 1+3

TOP 8/22 72 Ash, 73 Bh, 73 Ma, 74 Ma, 76 Bh, 78 Ch, 79 Bh, 81 Ba · **most repeated question in the chapter**

Definition

- A mobile moves out of one cell into another **while a call is in progress**.
- The MSC transfers the call to a new channel belonging to the new BS.
- That transfer = **handoff** (US) = **handover** (Europe).
- Must be **transparent**: the user should not notice it.

The handoff process, step by step

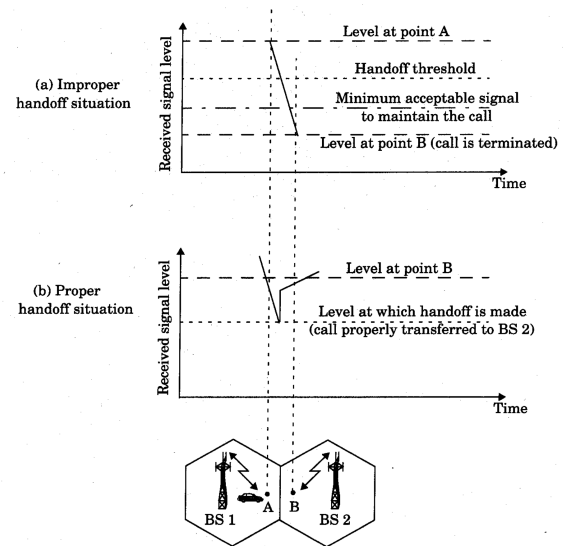
- Serving BS continuously measures the received signal strength on each **reverse voice channel** → tells the MSC roughly where each mobile is.
- Signal drops below the **handoff threshold** $P_{r,handoff}$.
- MSC **identifies a candidate BS** whose signal is stronger.
- MSC checks the candidate has a **free channel**.
- MSC **re-allocates voice and control channels** to the new BS.
- Call continues on the new channel, old channel returns to the pool.

Handoff threshold and handoff margin (75 Bh asks for the margin +Fig. 3)

- $P_{r,minimum\ usable}$ = weakest signal that still gives acceptable voice, typically -90 to -100 dBm.
- $P_{r,handoff}$ = level at which handoff is started. Always set a little higher.
- **Handoff margin**

$$\Delta = P_{r,handoff} - P_{r,minimum\ usable}$$

- **Δ too large** → handoffs fire needlessly → MSC is burdened.
- **Δ too small** → not enough time to finish the handoff → call drops.
- 1G analog: $\Delta \approx 6$ to 12 dB, handoff takes about 10 s.
- 2G (GSM): $\Delta \approx 0$ to 6 dB, handoff takes 1 to 2 s.



Improper Vs proper handoff (76 Bh 4, 81 Ba 1+3). Top: the level is left to fall past the minimum usable value before handoff is attempted → **call terminated at B**. Bottom: handoff is made while the signal is still usable → **call transferred to BS 2**.

Types of handoff

- **Hard handoff** = “**break before make**”.
 - Old connection is released **before** the new one is set up.
 - Used in FDMA and TDMA systems (AMPS, GSM), where the two cells are on different frequencies.
 - Risk: if the new channel fails, **the call drops**.
- **Soft handoff** = “**make before break**”.
 - New connection is set up **before** the old one is released, so the mobile is briefly served by two BSs at once.
 - Only possible in **CDMA**, where every cell uses the same frequency.
 - Gives **BS diversity** at the cell boundary → far fewer dropped calls.
- **Intra-cell handoff**: same BS, different channel. Used to escape interference or a deep fade.
- **Inter-cell handoff**: between two cells of the same MSC.
- **Intersystem handoff**: mobile leaves the coverage of one MSC and enters a system run by a **different MSC**. Needs an inter-MSC link and a roaming agreement.

How the handoff decision is taken (handoff measurement)

- **1G analog:** the **BS** measures signal strength, the **MSC** decides. A spare receiver, the **locator receiver**, scans neighbouring cells' mobiles.
- **2G TDMA: mobile assisted handoff (MAHO).**
 - The **mobile** measures received power from surrounding BSs and reports continuously to its serving BS.
 - Handoff triggers when a neighbour becomes appreciably stronger.
 - **Much faster (1 to 2 s)** because the MSC no longer has to poll every BS.
 - GSM also weighs **co-channel and adjacent channel interference**, not just level.
- **IS-95 CDMA:** all cells share the channel, so **there is no physical channel change at all**. The MSC simply picks the BS receiving the strongest signal.
- In every case the decision must use a **running average** of signal strength, so that **a momentary fade is not mistaken for the mobile leaving the cell**.
 - Averaging window depends on vehicle speed.
 - A steep short-term average → handoff must be made quickly.
 - Speed can be estimated from the statistics of the short-term fading at the BS.

Explain two schemes for prioritizing handoff 3 2

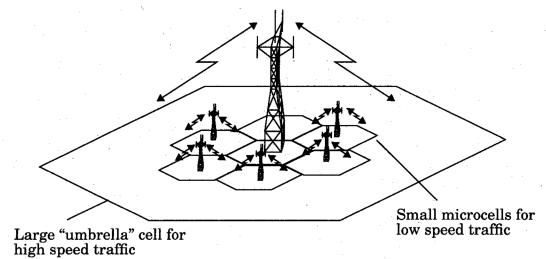
PIN 1/22 82 Bh · also asked inside 72 Ash 2

- **Why prioritize:** a **dropped call** is judged far more serious by the subscriber than a **blocked call**. So handoff requests must beat new-call requests.
- **Scheme 1: guard channels.**
 - A fixed fraction of each cell's channels is **reserved exclusively** for handoff requests.
 - Simple, immediate.
 - Cost: those channels sit idle for new calls → **total carried traffic falls**.
 - Fixed by using **dynamic channel assignment** for the reserved pool.
- **Scheme 2: queuing of handoff requests.**
 - A handoff request that finds no free channel is **put in a queue** instead of being refused.
 - Works because there is a **finite window** between the signal dropping below the handoff threshold and dropping below the minimum usable level.
 - That window leaves room to wait for a channel to free up.
 - Reduces the **probability of forced termination**, but does not eliminate it.

Practical handoff considerations 4 4+4

HOT 4/22 79 Bh, 73 Ma, 78 Ch, 72 Ash

- **Users move at wildly different speeds.**
 - High-speed vehicle users need **frequent** handoffs.
 - Pedestrians may need **none at all** in a whole call.
- **Microcells make it worse.** Small cells raise capacity, but the MSC is swamped when fast users are handed between them every few seconds.
- **Umbrella cell** is the standard fix +Fig.
 - A large, high, high-power cell is **overlaid** on a group of microcells at the same site.
 - **High-speed** traffic is put on the umbrella cell → few handoffs.
 - **Low-speed** traffic is put on the microcells → high capacity.
 - Speed is estimated from the rate of change of the short-term signal average.
- **Cell dragging** (72 Ash, 73 Bh ask for this by name, 2).
 - A **pedestrian** with a clear LOS to the BS keeps a **very strong** signal.
 - The level never falls to the handoff threshold, so no handoff is triggered.
 - The user **travels deep into a neighbouring cell** while still held by the old BS.
 - Result: **severe interference in the neighbouring cell, and the reuse plan is broken.**
 - Fixed by tuning the handoff threshold and averaging window carefully.
- **Handoff requests outrank new calls** so that GoS is preserved.



Umbrella cell: one large cell for fast traffic sitting over the microcells that serve slow traffic.

Handoff strategy used in GSM (70 Ma, 8)

- GSM uses **MAHO**, so the mobile does the measuring.
- In every idle timeslot the mobile measures the **BCCH** power of up to **16 neighbouring cells** and reports the best 6 to its BS twice per second.
- Decision uses **RXLEV** (level) and **RXQUAL** (bit error rate), not level alone.
- So GSM will hand a call off for **poor quality even when the level is high**, which a level-only 1G system cannot do.
- Handoff is **hard** (break before make) and completes in 1 to 2 s.
- Four cases: intra-cell, inter-cell within one BSC, between two BSCs of one MSC, and between two MSCs.

2.4 Interference and system capacity

Explain co-channel interference and adjacent channel interference 2.5+2.5 3 4

80 Ch, 70 Bh, 81 Bh, 80 Ba, 81 Ba · 70 Bh and 81 Bh pair it with the $Q = \sqrt{3N}$ proof

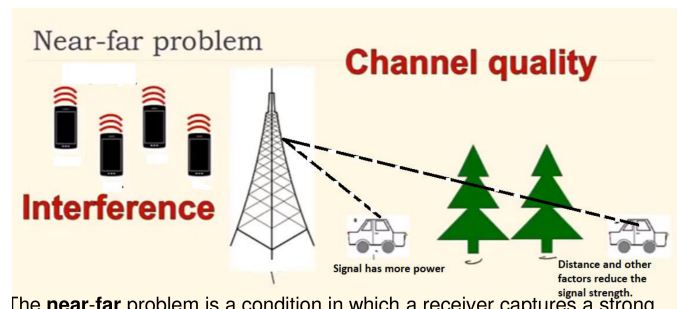
Sources of interference

- Another mobile **in the same cell**.
- A call in progress in a **neighbouring cell**.
- Other **base stations** operating in the same frequency band.
- Any **non-cellular system** leaking energy into the cellular band.
- Interference is **the limiting factor on capacity**, not noise.
- On **voice** channels it causes cross-talk and audible noise.
- On **control** channels it causes missed and blocked calls through signalling errors.

	Co-channel interference (CCI)	Adjacent channel interference (ACI)
Cause	Another cell using the same frequency	Signal in a nearby frequency leaking through an imperfect receiver filter
Comes from	Co-channel cells, one reuse distance away	Channels next to the wanted one, often inside the same cell
Depends on	$Q = D/R$ and cluster size N	Filter sharpness and the near-far geometry
Independent of	Transmit power (raising all powers raises S and I equally)	
Cured by	Increasing D/R , ie a larger N . Sectoring	Channel allocation w/ maximum frequency separation. Sharp band-pass filters. Power control

Adjacent channel interference in detail

- Receiver filters are not ideal → energy from neighbouring frequencies **leaks into the passband**.
- **Near-far problem** is the worst case + Fig.
 - A mobile **close** to the BS transmits on a channel adjacent to one used by a mobile **far** from the BS.
 - The near mobile's signal arrives far stronger.
 - The BS then **cannot hear the far mobile at all**.
- Cures:
 - **Channel allocation**: never give a cell channels that are adjacent in frequency. Keep separation as large as possible.
 - **Careful filtering**: sharp band-pass filter at the receiver, plus modulation w/ low out-of-band radiation.
 - **Separate multiplexing** for uplink and downlink.
 - **Power control** (below).



Power control syllabus, no PYQ yet

- Every mobile transmits **the smallest power** that still holds a good reverse link.
- Benefits:
 - **Solves the near-far problem**: the near mobile is turned down until the far mobile is audible again.
 - **Longer battery life**.
 - **Raises reverse-link SIR** across the whole system.
- **Essential, not optional, in CDMA**, where every user in every cell shares one radio channel, so one loud mobile can jam a whole cell.

Prove that the co-channel reuse ratio $Q = \sqrt{3N}$, and derive S/I

TOP 6/22 70 Bh, 71 Bh, 74 Ma, 79 Bh, 80 Bh, 81 Bh · almost always followed by “find optimal N for SIR = 15 or 18 dB” → Ch 2 numerical companion

Step 1: what Q is

- R = cell radius (centre to vertex).
- D = distance to the centre of the **nearest** co-channel cell.
- **Co-channel reuse ratio** $Q = D/R$.
- When all cells are about the same size, CCI is **independent of transmit power** and depends only on R and D .
- **Larger Q → less interference but less capacity.** The whole design is this trade-off.

Step 2: hexagonal geometry

- Put the cell centres on the axes u, v separated by 60° .
- Distance between two **adjacent** cell centres = $\sqrt{3}R$.
- Move i cells along u , then j cells along v at 60° .
- By the cosine rule w/ the included angle = 60° ,

$$D^2 = (i\sqrt{3}R)^2 + (j\sqrt{3}R)^2 - 2(i\sqrt{3}R)(j\sqrt{3}R)\cos 120^\circ$$

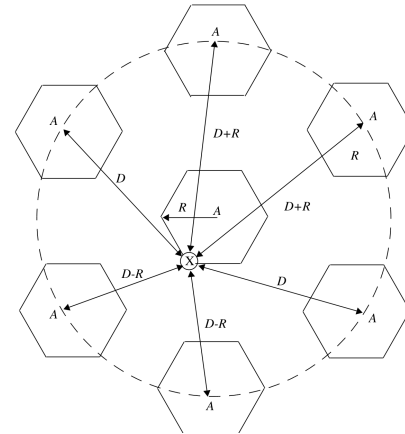
$$D^2 = 3R^2(i^2 + j^2 + ij)$$

- But $N = i^2 + ij + j^2$, so $D^2 = 3NR^2$, giving

$$D = \sqrt{3N}R \quad \Rightarrow \quad Q = \frac{D}{R} = \sqrt{3N}$$

Interference and System Capacity

71



First tier for $N = 7$. Worst case: the mobile sits at the cell boundary (point A), so its wanted signal is at its **minimum** while the interferers are at their nearest. Exact distances are $D - R$, D , D , D , $D + R$, but the approximation $D_i = D$ is used for easy analysis.

Step 3: signal to interference ratio

- Received power falls as $P_r \propto d^{-n}$, path loss exponent $n = 2$ to 4 .
- With N_I interfering cells,

$$\frac{S}{I} = \frac{S}{\sum_{i=1}^{N_I} I_i} = \frac{R^{-n}}{\sum_{i=1}^{N_I} D_i^{-n}}$$

- Take the **worst case**: mobile at the cell edge, first tier only, $N_I = 6$, all $D_i = D$:

$$\frac{S}{I} = \frac{R^{-n}}{6D^{-n}} = \frac{(D/R)^n}{6} = \frac{Q^n}{6} = \frac{(\sqrt{3N})^n}{6}$$

- Inverting for design,

$$Q = \sqrt{3N} = \left(6 \cdot \frac{S}{I}\right)^{1/n} \Rightarrow N = \frac{Q^2}{3}$$

- Then **round N up** to the next allowed $i^2 + ij + j^2$.

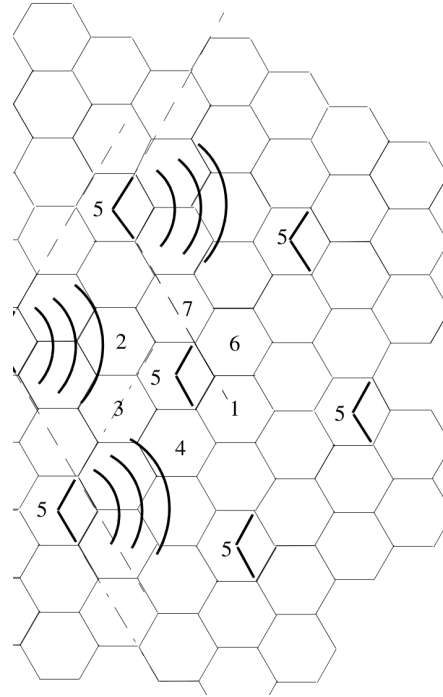
Exact worst case for $N = 7$, and what sectoring does to it

- Keeping the true distances instead of $D_i = D$, and $n = 4$:

$$\frac{S}{I} = \frac{R^{-4}}{2(D-R)^{-4} + 2(D+R)^{-4} + 2D^{-4}}$$

- For $N = 7$, $Q = 4.6$:
 - approximate result $\rightarrow 49.56 = 17$ dB,
 - exact result $\rightarrow 17.8$ dB.
- Both fall **just short of the 18 dB** that AMPS demands.
- Going to the next size, $N = 12$ ($i = j = 2$), fixes SIR but cuts spectrum utilisation from $1/7$ to $1/12 \rightarrow$ **not tolerable**.
- So the answer is **sectoring**, not a bigger N .

- With 120° sectoring and 7-cell reuse, a mobile is hit by only **2 of the 6** first-tier co-channel cells.
- SIR rises from 17 dB to about **24.2 dB**.
- That margin is then **spent** on reducing N , which raises capacity.



120° sectoring: of the 6 first-tier co-channel cells, only 2 have antenna patterns radiating into the centre cell.

2.5 Trunking and grade of service

Define grade of service. How is it measured in a blocked-calls-cleared trunking system? 3 4

PIN 1/22 79 Bh · 82 Bh asks for GoS, traffic intensity, average holding time and blocked call as definitions 4 · the numericals built on this are in the Ch 2 companion

What trunking is

- A large user population shares a **small pool of channels**.
- A channel is allocated **per call** and returned to the pool the moment the call ends.
- Works because users do not all call at once: it exploits their **statistical** behaviour.
- If every channel is busy when a user calls, that user is **blocked**.
- Trade-off: fewer channels → cheaper, but more blocking.

Definitions (82 Bh asks for exactly these)

- **Set-up time**: time to allocate a trunked channel to a requesting user.
- **Blocked call** (lost call): a call that cannot be completed at the time of request, because of congestion.
- **Holding time** H : average duration of a typical call, in seconds.
- **Request rate** λ : average number of call requests per unit time.
- **Traffic intensity** A : average channel occupancy, in **Erlangs**. Dimensionless.
- **Load**: traffic intensity across the whole system.
- **1 Erlang** = one channel occupied 100 % of the time (one call-hour per hour). Eg a channel busy for 30 min in an hour carries 0.5 Erlang.

The two types of trunked system

Blocked calls cleared → Erlang B

- **No queue**. A user who finds no free channel is refused and is free to try again.
- Assumptions: Poisson arrivals, infinite users, memoryless arrivals, exponentially distributed holding time, finite channels. This is an **M/M/m/m** queue.
- GoS = probability of blocking:

$$\Pr[\text{blocking}] = \frac{\frac{A^C}{C!}}{\sum_{k=0}^C \frac{A^k}{k!}} = \text{GoS}$$

- Gives a **conservative** (pessimistic) estimate, since a finite-user model always predicts less blocking.
- In the exam you are given an **Erlang B table or chart**, never asked to sum the series.

Grade of service (GoS)

- GoS = a measure of a user's ability to access the system **during the busy hour**.
- **Busy hour** = the busiest hour of a week, month or year. For cellular it is typically 4 to 6 pm Thursday or Friday.
- Stated either as
 - **probability a call is blocked** (Erlang B), or
 - **probability a call is delayed** beyond a set queuing time (Erlang C).
- AMPS is designed for **GoS = 2 % blocking**: 2 calls in 100 are lost in the busy hour.
- The designer's job: estimate peak demand, then allocate enough channels to meet the GoS.

The three traffic equations

- Per user: $A_u = \lambda H$
- Total offered, U users: $A = U A_u$
- Per channel, C channels: $A_c = U A_u / C$
- **Offered traffic is not carried traffic**. Once offered traffic exceeds capacity, the carried traffic saturates at C Erlangs.

Blocked calls delayed → Erlang C

- **A queue holds blocked calls** until a channel frees up.
- Probability a call is delayed at all:

$$\Pr[\text{delay} > 0] = \frac{A^C}{A^C + C! \left(1 - \frac{A}{C}\right) \sum_{k=0}^{C-1} \frac{A^k}{k!}}$$

- GoS = probability the delay exceeds t seconds:

$$\Pr[\text{delay} > t] = \Pr[\text{delay} > 0] \exp\left(-\frac{(C-A)t}{H}\right)$$

- Average delay over **all** calls:

$$D = \Pr[\text{delay} > 0] \frac{H}{C-A}$$

where $H/(C-A)$ is the average delay of *those* calls that were queued.

Trunking efficiency: a large trunked group carries far more traffic per channel than several small ones at the same GoS. This is why sectoring, which splits one channel pool into 3 or 6 smaller pools, always **loses** some trunking efficiency even though it improves SIR. (81 Ba asks for exactly this loss, 6)

2.6 Improving coverage and capacity

What techniques enhance capacity and coverage in a cellular radio network? 8 5 4

PIN 1/22 71 Ma · the sectoring Vs cell splitting comparison is HOT 3/22 73 Bh, 75 Bh, 81 Bh

Why it is needed

- A new system is deployed with few users, spread evenly → demand is low.
- As subscribers grow, demand in **some** cells exceeds what those BSs can carry.
- More spectrum is not available → capacity must come from **geometry**.

	Cell splitting	Sectoring	Microcell zone
Idea	Subdivide a congested cell into smaller cells	Replace the omni antenna w/ directional antennas	Split a cell into zones fed by one BS
R	Reduced	Unchanged	Unchanged
D/R	Unchanged	Reduced (via smaller N)	Reduced
Gains capacity by	More clusters per unit area	Allowing a smaller N	Allowing a smaller N w/o extra handoffs
Handoffs	More (smaller cells)	Many more (sector to sector)	None between zones
Cost	New BSs, towers, real estate	Loses trunking efficiency	Zone sites plus fibre or microwave links

Cell splitting 5

- **Subdivide** a congested cell into smaller cells, each w/ its **own BS** and **reduced** antenna height and transmit power.
- New smaller cells are called **microcells** and are dropped in between the existing cells.
- Halving R needs **about 4 times as many cells** to cover the same area ($\text{area} \propto R^2$).
- More cells → more clusters → more channels per unit area → higher capacity.
- **Cell splitting merely rescales the geometry**: $Q = D/R$ is unchanged, so the reuse plan still works.

The power reduction (commonly examined)

- Received power must be equal at the old and new cell edges, so the reuse plan behaves the same:

$$P_r[\text{old edge}] \propto P_{t1} R^{-n}, \quad P_r[\text{new edge}] \propto P_{t2} (R/2)^{-n}$$

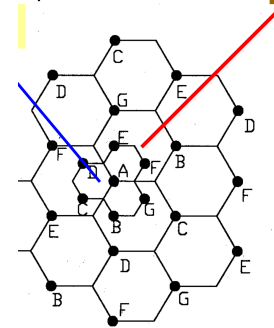
- Setting them equal w/ $n = 4$:

$$P_{t2} = \frac{P_{t1}}{16} \Rightarrow \boxed{\text{reduce transmit power by 12 dB}}$$

Problems

- In practice **not all cells are split at once** → two cell sizes coexist.
- Neither the old nor the new power works for both:
 - all-high power → small cells break the co-channel separation,
 - all-low power → parts of the large cells go unserved.
- So channels must be **split into two groups**, one per cell size. The large cell takes high-speed traffic so it hands off less often.
- Needs new towers and antennas, and **lowers spectral efficiency** during the transition.

reuse plan.
n power.



Reduce R to $R/2$

Each original cell is surrounded by microcells of half the radius, placed so the reuse plan is preserved.

Sectoring 5

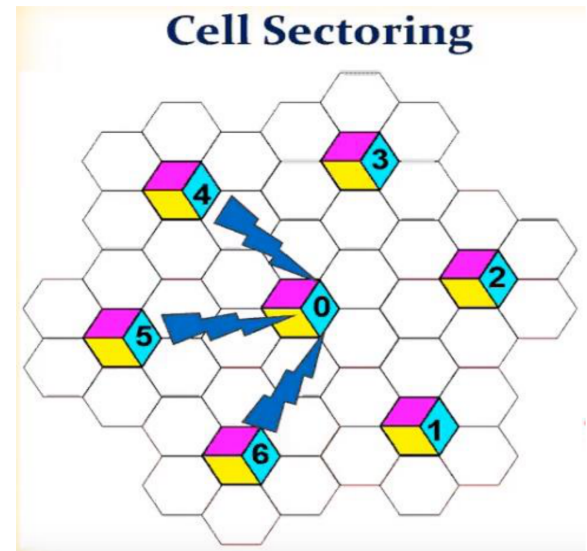
- Keep R fixed, and instead attack D/R .
- Replace the single omni-directional BS antenna w/ **several directional antennas**, each radiating into one **sector**.
- A cell is normally split into **three** 120° or **six** 60° sectors.
- Each sector gets its **own** channel sub-group, used only in that sector.
- A given cell now transmits to, and receives interference from, only a **fraction** of the co-channel cells.

The two-step gain (state it in this order)

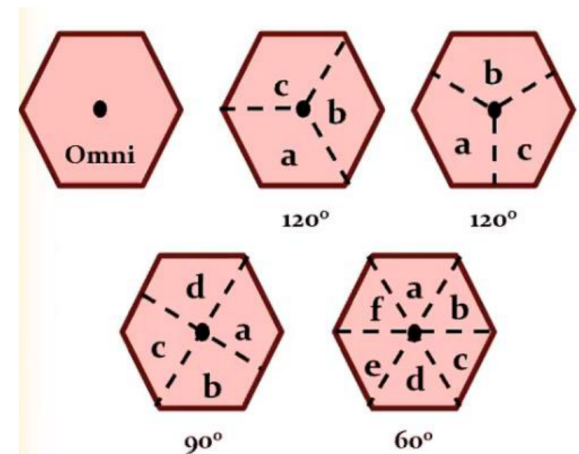
- **Step 1:** directional antennas cut the number of first-tier interferers.
 - omni $\rightarrow$ **6** interferers,
 - $120^\circ \rightarrow$ **2** interferers,
 - $60^\circ \rightarrow$ **1** interferer.
- SIR therefore **improves** ($17\text{ dB} \rightarrow$ about 24.2 dB for 120°).
- **Step 2:** spend that margin. Reduce N (eg $7 \rightarrow 4$ or 3), which raises capacity, until SIR is back at the required value.
- Note the transmit power is **not** reduced. Only the relative interference is.

Disadvantages

- The cell's channels are split into 3 or 6 smaller pools $\rightarrow$ **trunking efficiency falls**, so carried traffic drops for the same channel count.
- **Many more handoffs**, one per sector crossing, loading the MSC.
- More antennas per site.



Omni, 120° , 90° and 60° sectoring.



Sector 0 receives interference only from sectors 4, 5 and 6, so the 6 interfering cells drop to 3 (2 in the exact Rappaport treatment for 120°).

Repeaters for range extension syllabus, no PYQ yet

- Radio retransmitters used to cover **hard-to-reach places**: inside buildings, valleys, tunnels.
- **Bidirectional**: receives from the serving BS, amplifies, reradiates into the target area, and does the same in reverse.
- Works over the air $\rightarrow$ can be installed anywhere, repeats a whole band.
- Often feeds a **distributed antenna system (DAS)** inside a building.
- **A repeater adds coverage, not capacity.** It only reradiates the BS signal.
- It also reradiates the received **noise and interference**, so placement and gain settings need care.

Microcell zone concept 3 1+3

HOT 3/22 80 Ba, 81 Ba, 79 Ch · asked as “**why do we need it?**”, so lead with the problem

Why it is needed (the marked part)

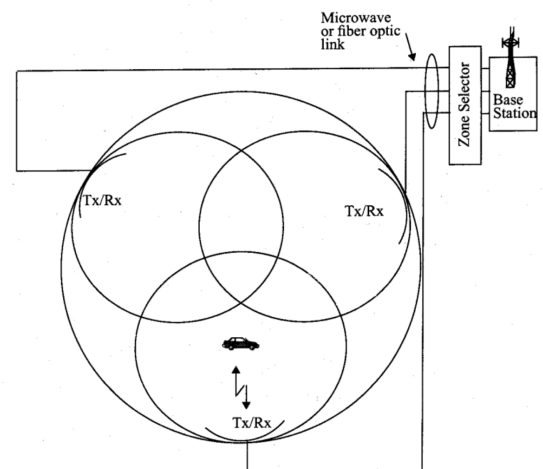
- Sectoring buys capacity but **multiplies handoffs**, one per sector boundary.
- Every handoff loads the MSC and the control links.
- Sectoring also **loses trunking efficiency** by splitting the channel pool.
- The microcell zone concept buys the same capacity **without either penalty**.

How it works

- A cell is divided into **3 or more zones**, each w/ a **zone site (Tx/Rx)** placed at the **outer edge** of the cell.
- All zone sites connect back to **one single BS**, by coaxial cable, fibre or microwave link, and share its radio equipment.
- Each zone uses a **directional antenna** radiating inward.
- The mobile is served by **whichever zone is strongest**.
- Any BS channel may be assigned to any zone.
- As the mobile crosses from zone to zone it **keeps the same channel**. **The BS just switches which zone site radiates it, so no handoff reaches the MSC.**

Result

- A channel is live only in the zone the mobile is in → radiation is localised → **interference falls**.
- D/R drops to about **3**, so N can drop from **7 to 3**.
- Capacity rises **about 2.33 times** at the same 18 dB SIR requirement.
- Particularly good along **highways and urban traffic corridors**.



Three zone sites (Tx/Rx), one base station, joined by fibre or microwave. A mobile crossing between zones keeps its channel.

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

Short	Means	Short	Means
BS	base station	BTS	base transceiver station
MS	mobile station	MSC	mobile switching centre
BSC	base station controller	MSK	minimum shift keying
CCI	co-channel interference	ACI	adjacent channel interference
SIR, S/I	signal to interference ratio	SNR, S/N	signal to noise ratio
GoS	grade of service	MAHO	mobile assisted handoff
FCA	fixed channel assignment	DCA	dynamic channel assignment
LOS	line of sight	NLOS	non line of sight
FDMA	frequency division multiple access	TDMA	time division multiple access
CDMA	code division multiple access	OFDM	orthogonal frequency division multiplexing
FDD	frequency division duplex	TDD	time division duplex
GSM	global system for mobile communications	AMPS	advanced mobile phone system
PCS	personal communication system	DAS	distributed antenna system
BCCH	broadcast control channel	EIRP	effective isotropic radiated power
ERP	effective radiated power	RF	radio frequency
Tx, TX	transmitter	Rx, RX	receiver
ISI	intersymbol interference	PDP	power delay profile
DFE	decision feedback equalizer	MLSE	maximum likelihood sequence estimation
LMS	least mean square	RLS	recursive least squares
MSE	mean square error	MMSE	minimum mean square error
ZF	zero forcing	MRC	maximal ratio combining
EGC	equal gain combining	BER	bit error rate
PCM	pulse code modulation	DPCM	differential PCM
ADPCM	adaptive differential PCM	LPC	linear predictive coder
RPE-LTP	regular pulse excited, long term prediction	CELP	code excited linear prediction
REL	residual excited linear prediction	MPE	multipulse excited
VSELP	vector sum excited linear prediction	ATC	adaptive transform coding
SBC	sub-band coding	DCT	discrete cosine transform
FEC	forward error correction	ARQ	automatic repeat request
BCH	Bose–Chaudhuri–Hocquenghem code	RS	Reed–Solomon code
TCM	trellis coded modulation	PN	pseudo-noise
MIMO	multiple input multiple output	OFDMA	orthogonal FDM access
QAM	quadrature amplitude modulation	PSK	phase shift keying
GMSK	Gaussian minimum shift keying	IMT	international mobile telecommunications
ASK	amplitude shift keying	FSK	frequency shift keying
BPSK	binary phase shift keying	QPSK	quadrature phase shift keying
OQPSK	offset (staggered) QPSK	DPSK,	differential (binary) PSK
		DBPSK	
CPFSK	continuous phase FSK	MFSK,	M-ary FSK, M-ary PSK
		MPSK	
DSSS	direct sequence spread spectrum	FHSS	frequency hopped spread spectrum
SS, SSM	spread spectrum (modulation)	PG	processing gain
PAPR	peak-to-average power ratio	CP	cyclic prefix
FFT, IFFT	(inverse) fast Fourier transform	ICI	inter-carrier interference
PSD	power spectral density	CFO, SFO	carrier, sampling frequency offset
NRZ, RZ	non-return, return to zero	AMI	alternate mark inversion
AM, FM, PM	amplitude, frequency, phase modulation	DSB, SSB,	double, single, vestigial sideband
		VS	
S/P, P/S	serial-parallel converter	D/A, A/D	digital-analog converter
LPF, BPF	low pass, band pass filter	ISM	industrial, scientific, medical band
SC-FDMA	single carrier FDMA	BT	GMSK filter bandwidth $\times$ symbol time
PSTN	public switched telephone network	VoIP	voice over IP
FVC, RVC	forward, reverse voice channel	FCC, RCC	forward, reverse control channel
WLAN	wireless local area network	WMAN	wireless metropolitan area network
IoT	internet of things	mmWave	millimetre wave
eMBB	enhanced mobile broadband	URLLC	ultra-reliable low latency comm.
mMTC	massive machine type comm.	NOMA	non-orthogonal multiple access
D2D	device to device	C-RAN	cloud radio access network
5G	5th generation mobile communication network	5G	5th generation mobile communication network

Standard and system names (proper names, kept as printed): **AMPS** advanced mobile phone system · **ETACS** extended total access communication system · **NMT** Nordic mobile telephone · **NTT** Nippon Telegraph and Telephone · **IMTS** improved mobile telephone service · **PDC** Pacific digital cellular · **IS-95** cdmaOne · **IS-136** digital AMPS · **HSCSD** high speed circuit switched data · **GPRS** general packet radio service · **EDGE** enhanced data rates for GSM evolution · **W-CDMA** wideband CDMA · **UMTS** universal mobile telecommunications system · **cdma2000 1xRTT / 1xEV-DO / 1xEV-DV** the cdma2000 3G family · **LTE, LTE-A** long term evolution (advanced) · **WiMAX** IEEE 802.16 · **WiFi** IEEE 802.11 · **DECT** digital enhanced cordless telecommunications · **CT2** cordless telephone 2 · **PHS** personal handyphone system · **PACS** personal access communications system · **POCSAG, FLEX** paging protocols · **CDPD** cellular digital packet data · **Mobitex** a dedicated packet data system · **TETRA** terrestrial trunked radio · **DAB** digital audio broadcasting · **DVB-T** digital video broadcasting, terrestrial · **ADSL** asymmetric digital subscriber line · **5G NR** 5G new radio.

One clash to watch: **FCC** is the *forward control channel* throughout these notes. The US regulator is always written out as the Federal Communications Commission.

Exam-paper month codes (used in the year tags): Ba = Baishakh · Bh = Bhadra · Ash = Ashwin · Ma = Magh · Ch = Chaitra.

Bold year tags are Regular exams, plain are Back exams. Monospace year tags are BEI (EX 715); the rest are BEX (EX 751).

Tier chips over 22 papers, 2070–2082 BS: **TOP /22** = topic asked in 6 or more papers · **HOT /22** = 3 to 5 papers · **PIN /22** = 1 or 2 papers but worth ≥ 6 marks. **Mark chips** such as **2+5** are the marks that question actually carried. **syllabus, no PYQ yet** marks a syllabus topic that has never been examined but is written up anyway. **verified/added** marks anything researched and added that was not in the source notes.